Volcano ECDLP census, Phase C: analysis and report

Study `research/volcano_ecdlp_census_20261017/`; plan `PLAN.md`, analysis plan `ANALYSIS_PLAN.md` (frozen after data collection and before inspection), amendments `AMENDMENTS.md` (A1, A2). Every number below is read from `analysis/results.json` (written by `analysis/analyze.py`, 10,000 bootstrap or permutation replicates, seeds 20261022 and 20261023). **Status: observed, not established.** No speedup is claimed. Independent validation (`VALIDATION.json`) has not happened yet; this report is the analysis, not its validation.

Erratum (A3, after independent validation)

The independent validator recomputed every number here (192 of 192 agree) and found two material reporting errors and six minor ones in the first version of this text; only wording changes, no estimand, margin, test, threshold or number. Corrected in place below and recorded in `AMENDMENTS.md` A3.

- **F1 (material).** The V3 j = 0 crater row of section 7 and the last paragraph of section 9 compared IC with `rho-neg`. On that curve the matched, strongest rho is `rho-cubic`; IC / `rho-cubic` is 84.9x / 92.8x / 131.1x at 0 / 0.1 / 1 GAE. The sentence that IC's gap to rho is smaller on that curve is withdrawn.
- **F2 (material).** The 'IC is 22 to 25% cheaper on the crater' observation and its automorphism explanation are withdrawn: the crater's own isomorphic re-scalings cost IC about 67 and 73 against 57.4 on the D/N model, so it is a single-Weierstrass-model effect, refuted as a level or j = 0 effect by the negative control.
- **F3.** `rho-cubic` ran on both re-scalings; 3 is the number of curve models, not generators. **F4.** The V2 `rho-plain` Ua level-0 cell averages 13.0 Velu evaluations; it is not 'no transport'. **F5.** Holm in section 5 is across 6 tests, not 10. **F6.** Two statements in `AMENDMENTS.md` A2 are corrected by A3. **F7.** 'about 43x' is 42.4x. **F8.** The V3 BSGS offset (1.014, 1.022) is a mix-of-scalars artifact. **N1.** See the limits.

1. Verdicts

1. **Positive control first (gates everything): PASS.** On the V3 (j = 0) crater, mean S(`rho-cubic`) / mean S(`rho-neg`) = **0.5733**, 95% CI [0.5619, 0.5850] over 4000 paired runs (theory 1/sqrt(3) = 0.5774; pass needs the CI upper bound below 0.85). The detector sees a 42% cost reduction, so the 'equivalent' verdicts below are informative, not relabelled 'uninformative'.
2. **Level effect (V1, V2; 5 arms; 35 level ratios): 33 equivalent, 0 different, 2 inconclusive.** TOST margins are +-5% for rho and BSGS and +-10% for IC. The two inconclusive ratios are both `rho-plain` at level 1 (CI not inside the margin; neither differs from 1 after Holm). No method shows a level effect on either volcano. With one volcano per l, this is an observed absence of an effect at these margins, not an established one (PLAN §5: replication on a second volcano of the same l is missing).
3. **Curve effect against the isomorphic negative control:** no variance-ratio test survives Holm (smallest adjusted p = 0.130); BSGS is degenerate (identical S for a curve and its re-scalings). 1 curve is flagged in total (V2, `rho-plain`, vertex 69), and it is not flagged for the other arms.
4. **Transfer:** every one of the 403168 measured runs recovered k (95% lower bound 0.9963 for the smallest cell). BSGS counts are identical across D, Ua, Ub and S in 100% of triples. Paired source ratios for rho-neg, rho-plain and ic: 23 of 24 equivalent, 1 inconclusive, 0 different. Transport is cheap against solving (section 6) and no 'transport, then solve' ratio is a speedup signal (0 signals).
5. **IC against matched `rho-neg` (distance remaining, not pass/fail):** IC costs 42.4x (V1) and 61.5x (V2) the rho cost in S with nothing priced, 65.1x and 92.7x with every uncharged operation priced at 1 GAE on both arms. The order (rho ahead) holds across the whole 0 / 0.1 / 1 band at every level of V1, V2 and V3 (on the V3 crater against its matched `rho-cubic`, A3/F1). No level improves IC's position.
6. **Wall time: not admitted for any (V, arm).** The A/A ratio is within [0.95, 1.05] on all three volcanoes, but only 87% to 94% of rows reached isolation L2, so condition 1 fails everywhere. Wall tables are descriptive.

2. What was measured

- 79 audited ecbench sessions (V3, V1, V2; V4 is censused, not measured, A1.1), each `audit OK` with exact replays; 403,168 measured runs after dropping 265,952 warm-up runs. Unverified measured runs: 0.
- Arms: `rho-neg` (matched reference), `rho-plain`, `bsgs`, `bsgs-neg`, `ic` (`ic-fb64-mitm`, frozen in `FROZEN.json`), `rho-neg-aa` (A/A), `rho-cubic` (V3 crater). Families D, Ua, Ub, S, N are isogenous; `ctl` (two isomorphic re-scalings per vertex) is the negative control and never pooled.
- Population: V1 (l = 2, depth 4, 400 curves per family), V2 (l = 3, depth 3, 378), V3 (j = 0, depth 4, 16; descriptive). Curves per level V1 25/25/50/100/200, V2 14/28/84/252, V3 1/1/2/4/8.
- Unit: S (ecbench normalised group operations). Host: shared 4-vCPU cloud VM; binary `ac4bc5b2...` from a clean commit (`runs/progress.jsonl` header); campaign wall about 12.2 h of sessions plus a 5-day pause.
- Statistics: cluster bootstrap over **vertices**, then rows, stratified by level (A2.3). Rows per curve are many; levels are compared on their curves, not their solves.

3. Positive control

V3 crater (census vertex 0), D, N and ctl rows pooled (3 curve models: D and N share the crater model, and `rho-cubic` ran on both re-scalings, 1,000 runs each), 4000 paired runs. Ratio 0.5733, 95% CI [0.5619, 0.5850] (pair bootstrap); slug-cluster sensitivity [0.5588, 0.5851]. Result: **pass**.

4. Level effect: R(L) = mean S at level L / mean S at the crater

Cells: ratio [95% CI], verdict (E equivalent, D different, I inconclusive). Margins [0.95, 1.053] for rho and BSGS, [0.90, 1.111] for IC. Holm family (a) has 35 tests.

V1 (l = 2)

arm	L1/L0	L2/L0	L3/L0	L4/L0	CV of S per row	achieved 95% half-width of R
rho-neg	0.992 [0.955, 1.032] E	0.997 [0.956, 1.040] E	1.006 [0.972, 1.043] E	1.002 [0.970, 1.036] E	0.46	0.037
rho-plain	1.016 [0.973, 1.062] I	0.997 [0.955, 1.042] E	0.987 [0.951, 1.026] E	0.994 [0.960, 1.030] E	0.47	0.040
bsgs	1.000 [0.991, 1.010] E	1.000 [0.991, 1.010] E	1.000 [0.992, 1.008] E	1.000 [0.993, 1.007] E	0.15	0.009
bsgs-neg	1.000 [0.986, 1.014] E	1.000 [0.986, 1.014] E	1.000 [0.988, 1.012] E	1.000 [0.989, 1.011] E	0.22	0.013
ic	0.980 [0.949, 1.012] E	0.981 [0.949, 1.015] E	0.984 [0.956, 1.013] E	1.011 [0.985, 1.038] E	0.35	0.030

V2 (l = 3)

arm	L1/L0	L2/L0	L3/L0	CV of S per row	achieved 95% half-width of R
rho-neg	0.993 [0.958, 1.030] E	0.984 [0.952, 1.016] E	0.998 [0.972, 1.025] E	0.46	0.032
rho-plain	1.030 [0.994, 1.069] I	1.007 [0.973, 1.044] E	1.019 [0.989, 1.051] E	0.46	0.035
bsgs	1.000 [0.989, 1.012] E	1.000 [0.990, 1.010] E	1.000 [0.991, 1.009] E	0.20	0.010
bsgs-neg	1.000 [0.983, 1.017] E	1.000 [0.985, 1.015] E	1.000 [0.987, 1.013] E	0.29	0.015
ic	0.990 [0.944, 1.038] E	0.996 [0.958, 1.036] E	0.983 [0.948, 1.019] E	0.45	0.040

BSGS ratios equal 1 exactly: BSGS cost depends on r and k only, and r is the same on every curve of a volcano. Their bootstrap interval reflects only the spread of the scalars k_i. Secondary mixed model (`log S ~ C(level) + C(family) + (1 | curve)`, REML) agrees: level ratios between 0.977 and 1.038; ICC between 0.0000 and 0.0074 (no curve-level variance beyond the solve noise); non-converged fits: V2|bsgs-neg (reported; the bootstrap is the result). Per-family R(L) (descriptive) are in `results.json` under `R_by_family_descriptive`.

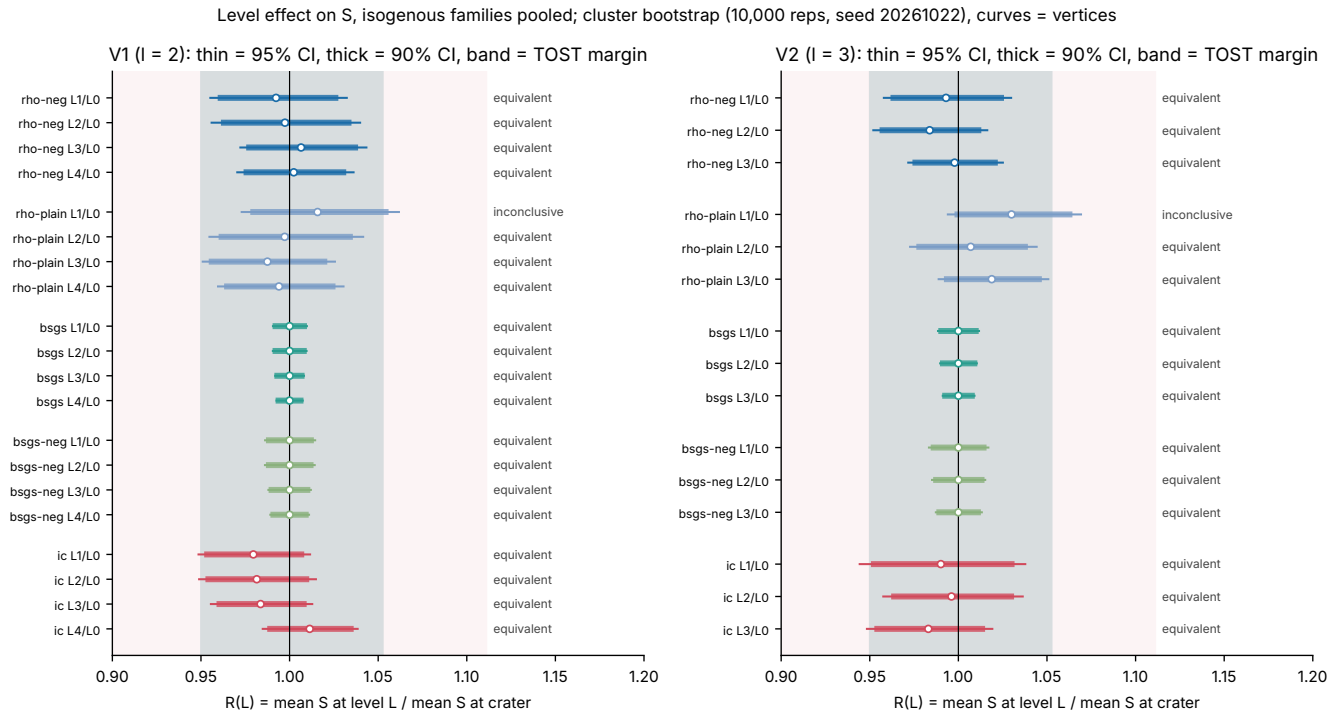


Figure 1. Level ratios with TOST bands. Source: `analysis/results.json` `level_effect`; session ids in `runs/progress.jsonl`.

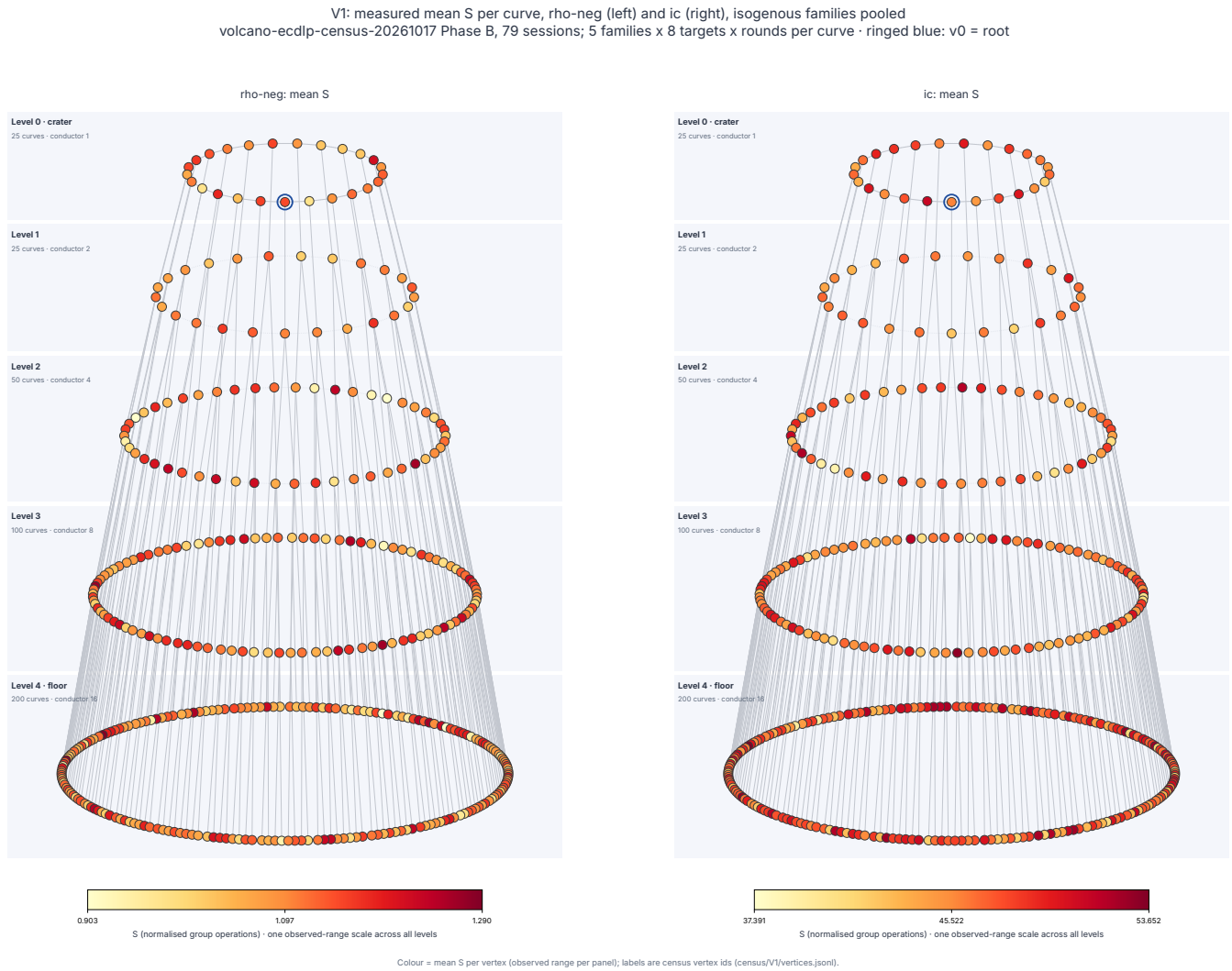


Figure 2a. V1 per-curve mean S for `rho-neg` and `ic` (standard `volcano-diagram/v2`). V2 and V3 are `fig2_volcano_V2/V3`. Colour scales are per panel; the spread within a level is the same size as between levels.

5. Curve effect against the negative control

volcano	arm	vertices	F (between / within)	permutation p	Holm p (across 6)	flagged curves (Holm p < 0.01)
V1	rho-neg	400	0.867	0.926	1.000	none
V1	rho-plain	400	1.032	0.676	1.000	none
V1	bsgs	400	degenerate	n/a	n/a	none (D and both re-scalings give identical S)
V1	bsgs-neg	400	degenerate	n/a	n/a	none (D and both re-scalings give identical S)
V1	ic	400	0.949	0.800	1.000	none
V2	rho-neg	378	0.900	0.888	1.000	none
V2	rho-plain	378	1.113	0.022	0.130	vertex 69 (level 2, z 4.55, adjusted p 1.0e-04)
V2	bsgs	378	degenerate	n/a	n/a	none (D and both re-scalings give identical S)
V2	bsgs-neg	378	degenerate	n/a	n/a	none (D and both re-scalings give identical S)
V2	ic	378	0.945	0.862	1.000	none

Holm is across the 6 non-degenerate (V, arm) tests; the 4 BSGS cells are degenerate and untested (A3/F5). F below 1 means the original curves spread no more than their own isomorphic re-scalings. The flagged V2 curve (vertex 69) has z by arm: rho-neg z = −0.24, rho-plain z = 4.55, ic z = 2.91. One curve of the 2,334 tested by the plan's rule (3 non-degenerate arms x 778 vertices), in one arm only, and the corresponding variance-ratio test is not significant after Holm; it is listed, not interpreted.

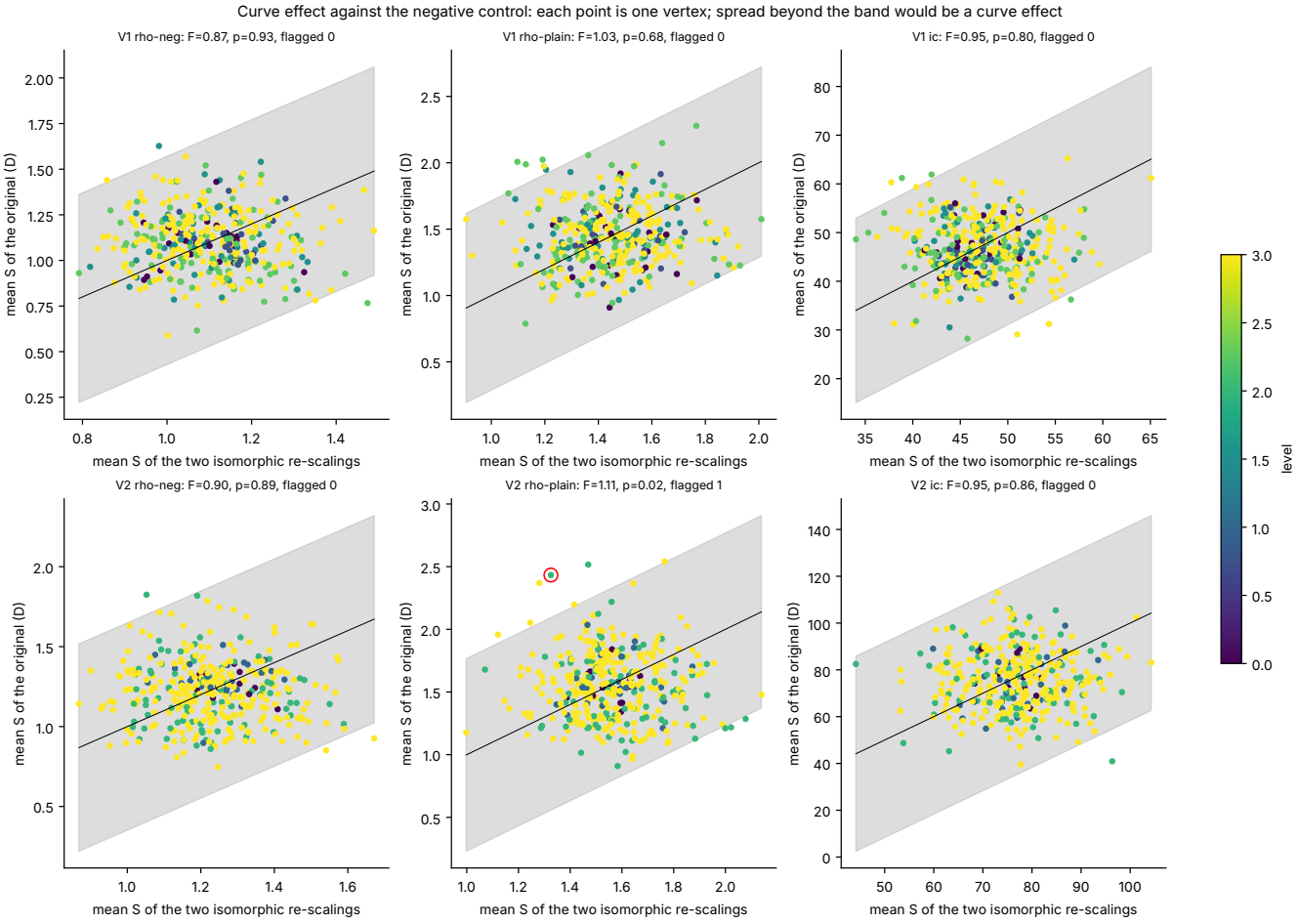


Figure 4. Mean S of each original curve (D) against the mean of its two re-scalings. The grey band is the normal-approximation 99% band of the control, for display only; flags come from the permutation test.

6. Transfer

Recovery. 403,168 of 403,168 measured runs verified (every (V, family, arm) cell 100%; the Clopper-Pearson 95% lower bound is 0.9963 for the 1,000-run cells and higher for larger ones). `matches_transported` is true for all 3,158 transport-benchmark paths.

Paired source ratios (mean S of the source family over D, on shared k_i; 95% CI; TOST margin as in section 4). Nothing is 'different'.

volcano	arm	Ua/D	Ub/D	S/D	N/D
V1	rho-neg	0.996 [0.969, 1.024] E	0.997 [0.969, 1.027] E	0.997 [0.969, 1.026] E	0.989 [0.962, 1.017] E
V1	rho-plain	1.018 [0.989, 1.049] E	0.992 [0.963, 1.022] E	1.028 [0.998, 1.059] I	1.021 [0.990, 1.052] E
V1	ic	1.010 [0.988, 1.033] E	1.002 [0.979, 1.026] E	0.991 [0.968, 1.015] E	0.997 [0.975, 1.019] E
V2	rho-neg	1.007 [0.978, 1.036] E	1.003 [0.976, 1.032] E	1.008 [0.979, 1.037] E	1.014 [0.985, 1.043] E
V2	rho-plain	0.995 [0.965, 1.025] E	1.004 [0.974, 1.034] E	1.003 [0.973, 1.034] E	1.001 [0.972, 1.031] E
V2	ic	0.999 [0.968, 1.030] E	1.015 [0.985, 1.047] E	1.001 [0.971, 1.031] E	1.017 [0.990, 1.046] E

N uses another scalar sequence (A1.3), so its pairing is nominal. BSGS identity check (fraction of (vertex, target, round) triples with equal `ops_gae` across the families present): V1 bsgs 1.000 over 3600; V1 bsgs-neg 1.000 over 3600; V2 bsgs 1.000 over 3584; V2 bsgs-neg 1.000 over 3584; V3 bsgs 1.000 over 2040; V3 bsgs-neg 1.000 over 2040.

Transport cost (`runs/transfer_cost.jsonl`, 200 repetitions per path, `ca volcano --transport-bench`; field-operation counts are a model, not instrumented). Per path length d, median ns per path (IQR) and Velu point maps (2 per edge):

volcano	d = 1	d = 4	d = 8	d = max
V1	2.8 us (2.7 to 2.8), d = 1, 2 maps	11.3 us (11.0 to 12.2), d = 4, 8 maps	22.3 us (22.1 to 22.7), d = 8, 16 maps	60.3 us (56.6 to 62.9), d = 20, 40 maps
V2	3.1 us (3.0 to 3.1), d = 1, 2 maps	12.5 us (12.3 to 12.7), d = 4, 8 maps	25.3 us (24.7 to 27.6), d = 8, 16 maps	44.7 us (40.4 to 45.3), d = 13, 26 maps
V3	3.2 us (3.2 to 3.3), d = 1, 2 maps	12.8 us (12.3 to 13.1), d = 4, 8 maps	n/a	19.7 us (19.0 to 20.3), d = 6, 12 maps

Transport as a fraction of the destination solve (median `solve_wall_ns` of the arm at that level; wall is descriptive, section 8):

volcano	arm	median fraction over levels (max over paths)	Velu maps / mean ops_gae at 0.1 / 1 GAE per map
V1	rho-neg	0.35% (1.03%)	0.055% / 0.55%
V1	bsgs-neg	0.38% (1.12%)	0.058% / 0.58%
V1	ic	0.06% (0.18%)	0.001% / 0.01%
V2	rho-neg	0.27% (1.21%)	0.092% / 0.92%
V2	bsgs-neg	0.29% (1.29%)	0.102% / 1.02%
V2	ic	0.09% (0.39%)	0.001% / 0.01%

Transport, then solve, against solve where born (A2.3; per source family, arm and destination level, priced at 0 / 0.1 / 1 GAE per Velu map): 240 cells; the CI excludes 1 at all three prices in 14 of them, and **0 meet the plan's speedup condition** (excludes 1 on the lower side at all three prices and disjoint from the native-instance control). The cells that exclude 1 are V1 `ic` from S (ratio about 1.2, above 1), V3 `ic` from D and Ua (about 1.2, above 1; a single-Weierstrass-model effect, section 9) and V2 `rho-plain` from Ua (about 0.6, **below 1**). The V2 `rho-plain` Ua cell is about 0.6 at every level, although the mean Velu evaluations per path rise from 13.0 (level 0, reached by transport from floor source vertex 364) to 18.9 (A3/F4; only rows at the source vertex itself have no transport), and the paired Ua/D ratio for it is equivalent to 1, so it reflects a high mean at the single source vertex (one floor curve, few rows), not a transport benefit; its CI overlaps the native-instance control CI (about 0.75 to 1.34), so it is not a signal under the plan's condition. The price band moves any ratio by at most 1.5%. Transport is at most about 1.3% of a solve in wall time for any path (medians below 0.5%) and at most about 1.3% in GAE at 1 GAE per Velu map (V2, `bsgs-neg`), far less for IC.

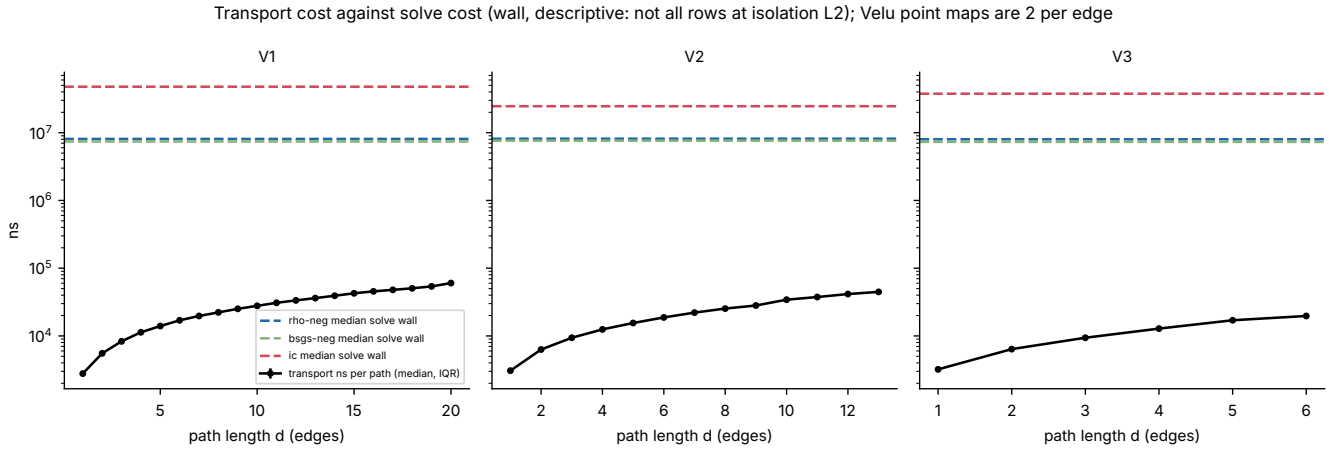


Figure 5. Transport ns per path against median solve wall by path length (descriptive wall).

7. IC against matched rho (distance remaining)

Mean S(ic) / mean S(rho-neg), isogenous families pooled, 95% CI, with every uncharged native operation (`ic`: lookups, target-guard probes, row operations, Legendre symbols, square roots; `rho-neg`: canonicalisations) priced at 0 / 0.1 / 1 GAE on **both** arms. Counts come from the run records (A2.1). Both arms still omit native work ecbench does not name, so both are lower bounds.

V1

level	curves	price 0	price 0.1	price 1	order across the band
0	25	42.6 [41.0, 44.4]	46.6 [44.8, 48.5]	65.4 [62.8, 68.3]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
1	25	42.1 [40.8, 43.4]	46.0 [44.6, 47.4]	64.6 [62.5, 66.7]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
2	50	41.9 [40.3, 43.6]	45.8 [44.1, 47.6]	64.4 [61.8, 67.0]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
3	100	41.6 [40.6, 42.6]	45.5 [44.4, 46.6]	63.9 [62.3, 65.5]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
4	200	43.0 [42.2, 43.8]	47.0 [46.2, 47.8]	66.0 [64.8, 67.2]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
all		42.4 [41.9, 42.9]	46.4 [45.8, 46.9]	65.1 [64.3, 65.9]	

V2

level	curves	price 0	price 0.1	price 1	order across the band
0	14	61.9 [59.4, 64.5]	67.3 [64.6, 70.2]	93.3 [89.3, 97.3]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
1	28	61.7 [59.1, 64.5]	67.1 [64.3, 70.2]	93.0 [88.9, 97.4]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
2	84	62.7 [60.7, 64.7]	68.2 [66.0, 70.4]	94.5 [91.4, 97.7]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)

3	252	61.0 [59.9, 62.2]	66.3 [65.1, 67.6]	91.9 [90.1, 93.7]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
all		61.5 [60.6, 62.4]	66.9 [65.9, 67.9]	92.7 [91.3, 94.1]	

V3 (descriptive: one curve at levels 0 and 1, 2 / 4 / 8 above; reference `rho-neg` unless stated)

level	curves	price 0	price 0.1	price 1	order across the band
0	1	49.0 [47.7, 50.3]	53.3 [51.9, 54.7]	73.3 [71.3, 75.3]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
1	1	59.9 [58.3, 61.5]	65.2 [63.4, 67.0]	90.1 [87.6, 92.7]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
2	2	62.6 [59.4, 66.0]	68.2 [64.6, 71.9]	94.4 [89.3, 99.6]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
3	4	62.0 [60.0, 64.0]	67.5 [65.3, 69.6]	93.3 [90.3, 96.4]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
4	8	60.5 [58.6, 62.4]	65.8 [63.8, 68.0]	91.0 [88.1, 94.0]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
0, matched rho = `rho-cubic` (D + N pairs, 2000)	1	84.9 [82.7, 87.0]	92.8 [90.4, 95.2]	131.1 [127.6, 134.6]	rho-cubic ahead of ic at all three prices
all		58.8 [57.8, 59.8]	64.0 [62.9, 65.1]	88.4 [86.9, 89.9]	

Uncharged operations per run (V1): `ic` 292,972 (about 1.9 per charged GAE), `rho-neg` 3,239 (about 0.90 per charged GAE). The gap to rho is therefore 42.4x unpriced and 65.1x at 1 GAE (V1), a distance of more than an order of magnitude at this size and no level of any volcano changes it. This is the measured baseline gap, not a result for or against IC's potential.

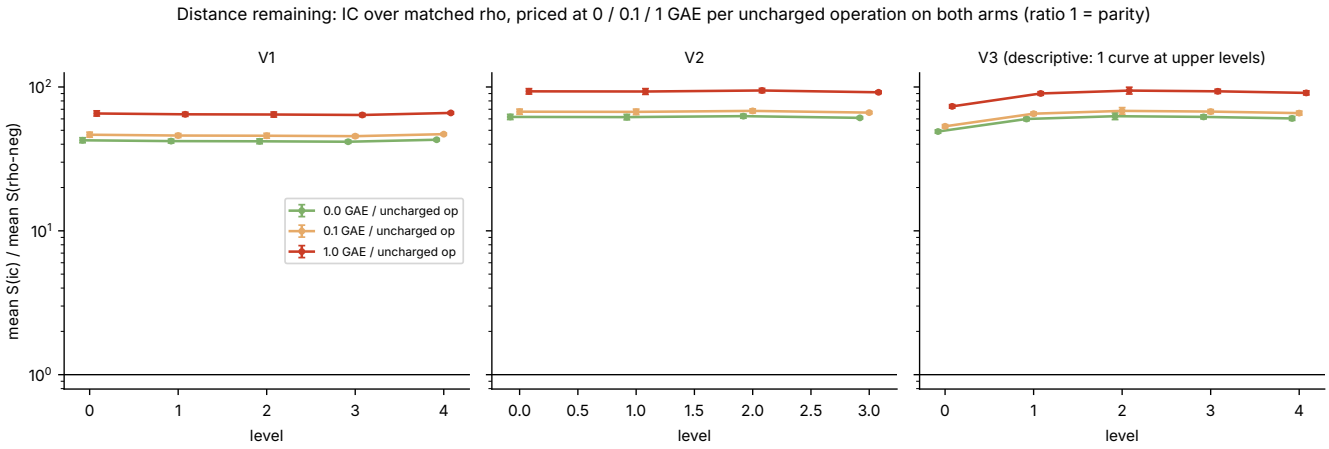


Figure 3. IC / rho-neg by level, three prices, 95% CI (log scale).

8. Wall time

volcano	A/A ratio (rho-neg-aa / rho-neg) [95% CI]	within [0.95, 1.05]	rows at L2 (rho-neg / ic)	admitted
V1	0.9994 [0.9944, 1.0044]	yes	92.0% / 86.8%	no (isolation L1 rows present)
V2	1.0005 [0.9959, 1.0051]	yes	93.4% / 91.6%	no (isolation L1 rows present)
V3	0.9978 [0.9924, 1.0032]	yes	92.4% / 89.4%	no (isolation L1 rows present)

So wall numbers (`solve_wall_ns` ratios in `results.json`, `R.<level>.wall`) are **descriptive (not admitted)**. For orientation they also show no level trend. The A/A result says the instrument's own noise floor is within 1% pooled.

9. V3 (j = 0), descriptive

One curve at levels 0 and 1, so no between-curve inference. Mean S relative to the crater:

arm	L1	L2	L3	L4
rho-neg	1.003	0.969	0.992	1.003
rho-plain	0.991	0.980	0.997	0.994
bsgs	1.014	1.014	1.014	1.014
bsgs-neg	1.022	1.022	1.022	1.022
ic	1.226	1.238	1.254	1.237

Observed, and refuted as a level or j = 0 effect by the study's own negative control (A3/F2): `ic` is about 22 to 25% more expensive at every non-crater V3 level than on the crater's D and N model. The crater's two isomorphic re-scalings, which have the same curve and the same automorphism group, cost `ic` 67.0 +- 0.9 (SE, 1000 rows, models a41c0795, ctl) and 72.6 +- 1.0 (SE, 1000 rows, models d53b367d, ctl) against 57.4 +- 0.5 on the D/N model (eeb6174a), in line with the non-crater levels (about 70 to 72). So this is an effect of one Weierstrass model (a coordinate effect of the prime-abscissa factor base), not of the level or of j = 0. The earlier automorphism explanation is withdrawn. The same model drives the V3 D and Ua 'transport, then solve' cells of about 1.2. BSGS shows a constant 1.014 (`bsgs`) and 1.022 (`bsgs-neg`) off the crater: this is a mix-of-scalars artifact, not a level effect. Crater rows are D + N (N uses another scalar seed, mean S(N)/S(D) = 0.946), other levels are D + Ua + Ub + N, and (3 + 0.946)/4 / ((1 + 0.946)/2) = 1.0139 (A3/F8).

IC against the matched rho on the V3 crater (A3/F1). On the j = 0 crater the strongest available rho is the automorphism-folded `rho-cubic`, not `rho-neg`. Against it IC costs 84.9x / 92.8x / 131.1x at 0 / 0.1 / 1 GAE (0 GAE 95% CI [82.7, 87.0]), larger than the gap against `rho-neg` at the non-crater V3 levels (about 60x / 65x / 90x). The earlier statement that IC's gap is smaller on that curve is withdrawn.

10. CV and achieved precision

CV and achieved precision (mean over levels); BSGS is deterministic in r and k, so its CV is the scalar's spread only

volcano	arm	curves per level	rows	CV of S per row	95% half-width of level mean	95% half-width of R(L)	ICC (mixed model)	converged
V1	rho-neg	25/25/50/100/200	18000	0.46	2.3%	0.037	0.000	yes
V1	rho-plain	25/25/50/100/200	18000	0.47	2.5%	0.040	0.000	yes
V1	bsgs	25/25/50/100/200	18000	0.15	0.6%	0.009	0.000	yes
V1	bsgs-neg	25/25/50/100/200	18000	0.22	0.8%	0.013	0.000	yes
V1	ic	25/25/50/100/200	18000	0.35	1.9%	0.030	0.004	yes
V2	rho-neg	14/28/84/252	17920	0.46	2.2%	0.032	0.002	yes
V2	rho-plain	14/28/84/252	17920	0.46	2.2%	0.035	0.000	yes
V2	bsgs	14/28/84/252	17920	0.20	0.7%	0.010	0.000	yes
V2	bsgs-neg	14/28/84/252	17920	0.29	1.0%	0.015	0.000	no
V2	ic	14/28/84/252	17920	0.45	2.5%	0.040	0.007	yes

Figure 6. Per (volcano, arm): curves per level, rows, per-row CV of S, achieved 95% half-width of the level mean and of R(L), and ICC. Rho CV about 0.46 (calibration 0.40, plan 0.52), IC 0.35 to 0.45, BSGS 0.15 to 0.29. The achieved 95% half-width of R(L) is 3 to 4% for rho and IC and about 1% for BSGS, so the +-5% rho margin and the +-10% IC margin were within reach, as the `rho-plain` level-1 results show (they sit just outside).

11. Limits

- Shared 4-vCPU cloud VM; wall time is descriptive, 8 to 13% of rows were at isolation L1.
- One volcano per l (V1 l = 2, V2 l = 3): no replication on a second volcano of the same l, so nothing can be 'established', only observed.
- V3 is descriptive (one curve per upper level, A1.2); its positive control pools D, N and both ctl re-scalings (4,000 pairs).
- V4 (twist of V1) is censused, not measured (A1.1).
- 5-day pause in Phase B (driver died after V2-S-L3-c0, resumed 2026-10-22); the gap is excluded from wall; no result depends on it.
- A1 amendments (V4, V3 shape, target seeds, rho CV, Drive, storage, clean binary) and A2 amendments (this phase, post hoc) shape the analysis; each interpretation is listed in `AMENDMENTS.md`. `ANALYSIS_PLAN.md` was frozen after data collection but before anyone inspected a Phase B cost summary. A 100-replicate smoke run of the code was inspected before the final run. `results.json` records the SHA-256 of `AMENDMENTS.md` as it stood at the final run, before A2.3 and A2.4 were appended.
- Curves are census vertices; the five isogenous families share a vertex and are pooled. A different clustering would change widths, not point estimates.
- IC and rho both leave unnamed native work unpriced; the 0 / 0.1 / 1 GAE band covers only counters ecbench names. Field-operation counts for Velu maps are a model.
- Small sizes only (26-bit group orders); no statement about cryptographic sizes, no speedup, no `claim`.
- Methodological note N1 (validator): the frozen two-stage bootstrap (resample curves, then rows within each curve) counts within-curve variance twice when the ICC is near 0 (mixed-model ICC at most 0.0074), so its CIs are conservative by about sqrt(2). This pushes knife-edge cells toward 'inconclusive' (V1 and V2 `rho-plain` L1/L0, V1 `rho-plain` S/D). It is the pre-registered procedure and no verdict changes; read those 'inconclusive' cells as a feature of the procedure, not only of the data's precision.
- The analysis is Python (the plan names it); AGENTS.md prefers native code for results. The validator should re-derive the headline numbers independently.

12. Files

- Code: `analysis/analyze.py`, `analysis/extract_uncharged.py`, `analysis/figures.py`, `analysis/build_report.py`; output `analysis/results.json` (all statistics, n, seeds, SHA-256 of every input), `analysis/figures/*.svg|png`, `analysis/analyze.log`.
- Inputs: `runs/extract/*.csv.xz`, `runs/uncharged/*.csv.xz`, `runs/slugmap/*.json`, `census/V*/`, `runs/transfer_cost.jsonl`, `runs/progress.jsonl`, sessions archives `sessions/*.tar.xz`.

- Copies for browsing: `benchmarks/20261017-volcano-ecdlp-census/` (no `benchmarks/README.md` row until the validation gate is passed).